

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Executed installation, decommissioning, and preventative maintenance for electromechanical semiconductor equipment at a key customer site, ensuring minimal disruption to their production schedule.
- Performed detailed troubleshooting and precise problem-solving on advanced semiconductor manufacturing tools, restoring optimal performance and reducing mean time to repair by **18%**.
- Provided timely updates and clear technical solutions to customer engineering teams, maintaining high levels of satisfaction and reinforcing long-term trust in service delivery.
- Collaborated with regional and global engineering teams to escalate complex issues, share operational insights, and contribute to the continuous improvement of service protocols.
- Adhered to strict safety and quality protocols within a cleanroom environment, ensuring all service activities met stringent equipment and operational standards.
- Developed Python scripts to automate data extraction and reporting from production logs, providing actionable insights that supported proactive maintenance scheduling for **50+** equipment units.
- Containerized Java Spring Boot microservices using **Docker** and deployed them to an **AWS EC2** cluster, streamlining the build and deployment process.

SKILLS

Semiconductor Equipment – Installation, Decommissioning, Preventative Maintenance, Repair, Troubleshooting, Electromechanical Systems, Semiconductor Wafer Fab Operations

Programming Languages – Java, Python, SQL, Shell Scripting

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Technologies – Splunk, Dynatrace, Jira, Confluence, Linux, R, RStudio, Power BI, Tableau, DAX, Power Query, Microprocessor Controllers, Electropneumatic Controls, Vacuum and Gas Control Systems

Engineering Principles – Electrical Schematics, Gas Schematics, Mechanical Troubleshooting, Electrical Troubleshooting, Robotics, Handling of Toxic/Flammable Gases and Chemicals

PROJECTS

Automated Production Monitoring System (HCL Technologies)

- Engineered a Python-based system to ingest and analyze production logs from multiple semiconductor manufacturing tools, identifying recurring error patterns.
- Developed **dashboard visualizations** in Power BI to present key performance indicators and failure trends to operations teams, enabling faster root cause analysis.
- Implemented automated alerts via email and Slack for critical system failures, reducing response time for P1 incidents by an average of **30 minutes**.

Equipment Performance Analytics Tool (MSc Data Analytics Project)

- Designed and implemented a data pipeline using Python (pandas, SQL) to aggregate historical performance data for a fleet of **100+** semiconductor processing machines.
- Developed a predictive maintenance model using scikit-learn to forecast potential equipment failures based on sensor readings and operational history, achieving **~0.82 AUC** on validation data.
- Created interactive reports and visualizations in Tableau to allow engineers to explore performance metrics, identify bottlenecks, and optimize maintenance schedules.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)